



Storage Management

Storage--> Storage Management in Linux refers to how data is stored, accessed, and organized in a Linux-based system. It includes managing hard disks, partitions, file systems, and mount points.

Storage Type

- 1. Floppy disk**
 - 2. CD/DVD**
 - 3. Pendrive/Memory card**
 - 4. Harddisk --> IDE,SATA,SSD,Nvme**
- Disk Partitioning:-** Disk partitioning allows system administrators a hard drive into multipal logical storage units, referred to as partitions.
 - -->** The process of dividing a disk into logical areas that can worked with separately is called partitioning.

- **Type of Partition Scheme**

1. MBR

2. GPT

MBR Partition Scheme	GPT Partition Scheme
* max. 4 primary partition * Max. 15 partition(logical+ext) * Max. Partition size (2TB) * system running BIOS based	* Max. 128 partition * Size Bzebibyte (8 million TB) * GPT store information in header & end of disk also * system running UEFI based

- **Commands**

1. fdisk --> GPT/MBR(create partition in mbr/gpt)

2. gdisk --> GPT (create partiton in GPT)

3. parted --> GPT/MBR(create partition in mbr/gpt)

4. lsblk --> show harddisk and storage

5. blkid --> shows info like UUID, type of storage devices

6. Df -Th --> shoe disk usage + file system type in GB/Mb

7. mkfs --> make file sysytem

8. mount --> attach (mount) storage device to directory

9. swapon --> enable or activie swap memory

10. swapoff--> disable or deactive swap memory

11. free -h --> show swap memory

12. pvcreate --> create a PV (physical volume)

13. **vgcreate --> create VG (volume group)**
14. **lvcreate --> create a logical volume**
15. **pvdisplay --> shows detailed information about a Physical Volume (PV).**
16. **vgdisplay --> shows detailed information about a volume group (VG)**
17. **vgs --> check vg**
18. **pvs --> check pv**
19. **lvs --> check lv**
20. **lvremove --> delete logical volume**
21. **vgremove --> delete volume group**
22. **pvremove --> delete physical volume**
23. **umount --> detached storage device and directory**

- **/dev = /dev stands for "device".**

It is a special directory in Linux that contains device files, which represent hardware devices connected to your system.

Partition with MBR

Attach a hard disk (/dev/sda)

```
fdisk /dev/sda
press (n) --> new partition
select (p) --> for primary partition
select P.no --> Partition number
frist sector --> enter
last sector --> size of partition (+1G , +2G)
press (p) --> for verifiy/ print
press (w) --> for write and save
```

Formate the partition with the “File System”.

File syystem types = ext2,ext3,ext4,xfs,vfat,btrfs

mkfs.ext4 /dev/sda1

mount /dev/sda1 /mnt

--> Do temporary mounting with the mount command.

--> Permanent mounting “/etc/fstab file

Partition with GPT

```
fdisk /dev/sda1
```

press (g) --> to create GPT type partition

select partition number --> 1 to 128

First sector --> enter

last sector --> +2G

Press (p) --> for print and verify

Press (w) --> to quit and save

```
mkfs.ext4 /dev/sda1
```

```
mount /dev/sda1 /mnt
```

* Permanent mounting

--> open the file /etc/fstab

```
--> /dev/sda1/  /mnt  ext4  default  0  0
```

--> then run command = mount -a

Swap Memory

Swap memory is a special area on the disk that is used as virtual memory when the physical RAM (Random Access Memory) is full.

Swapiness value = 60

free -h = show swap memory

Add a Swap Memory

```
STEP:
Add new Partition

fdisk /dev/sdb
press --> n
select partition number --> 1,2,3
first sector --> enter
last sector --> +1G

** Now we need to change the partition type
press --> t
press (l) --> to list all types
--> search for swap partition
--> for gpt select 19
--> for mbr select 82
press(w) --> save

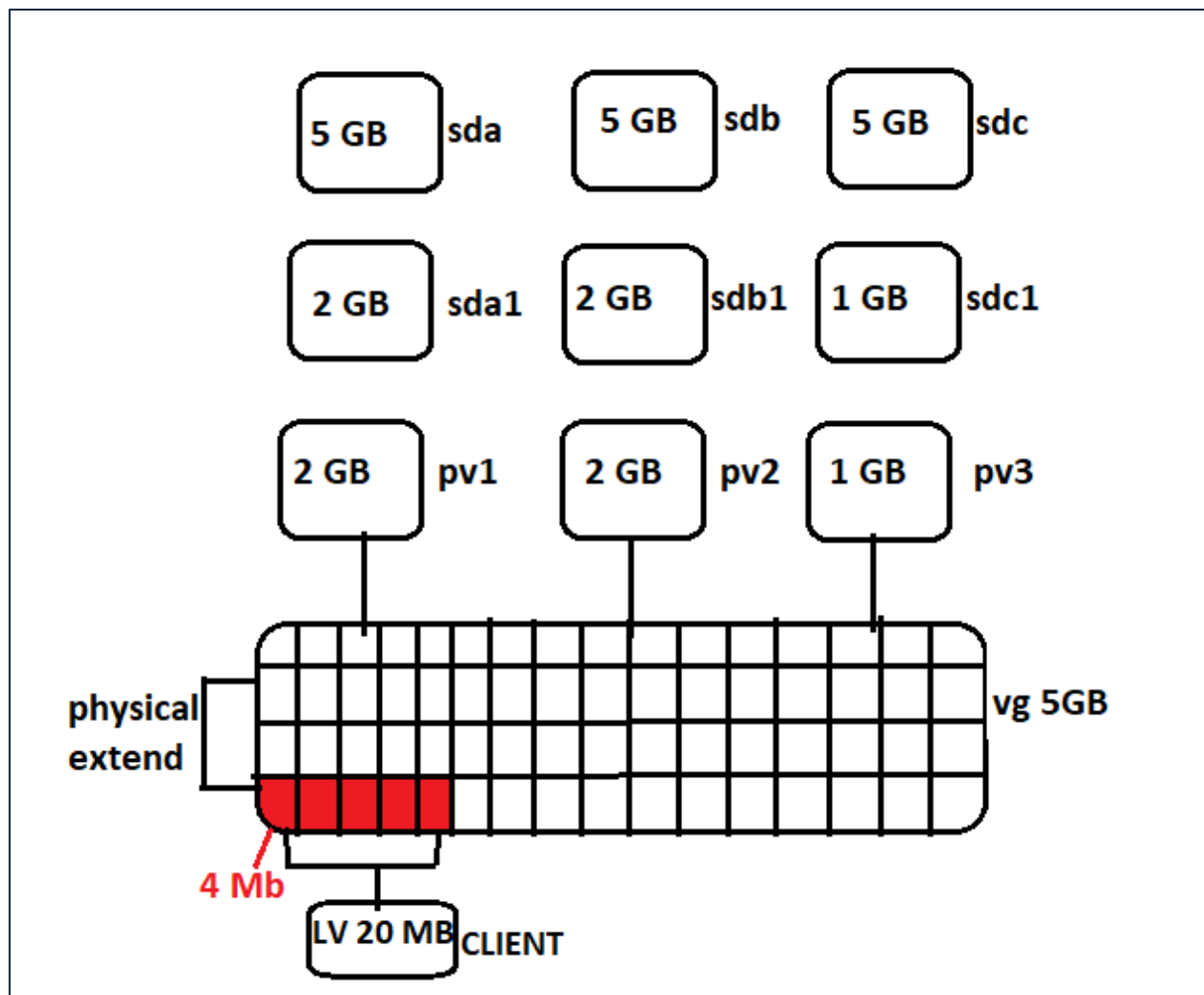
formate --> mkswap /dev/sdb1
--> swapon /dev/sdb1

Parmanent mounting for Swap
--> vim /etc/fstab
--> /dev/sdb1 swap swap defaults 0 0
```

LVM (Logical Volume)

LVM (Logical Volume Manager) is a flexible storage management system in Linux.

It allows you to manage disk space more dynamically and easily than traditional partitioning.



Create LVM

fdisk /dev/sda

Press --> n

Select partition number --> 1

Frist sector --> enter

Last sector --> +3G

**** now we need to change the partition type**

Press --> t

Press(l) --> to list all type (lvm -8e)

press (w) --> save

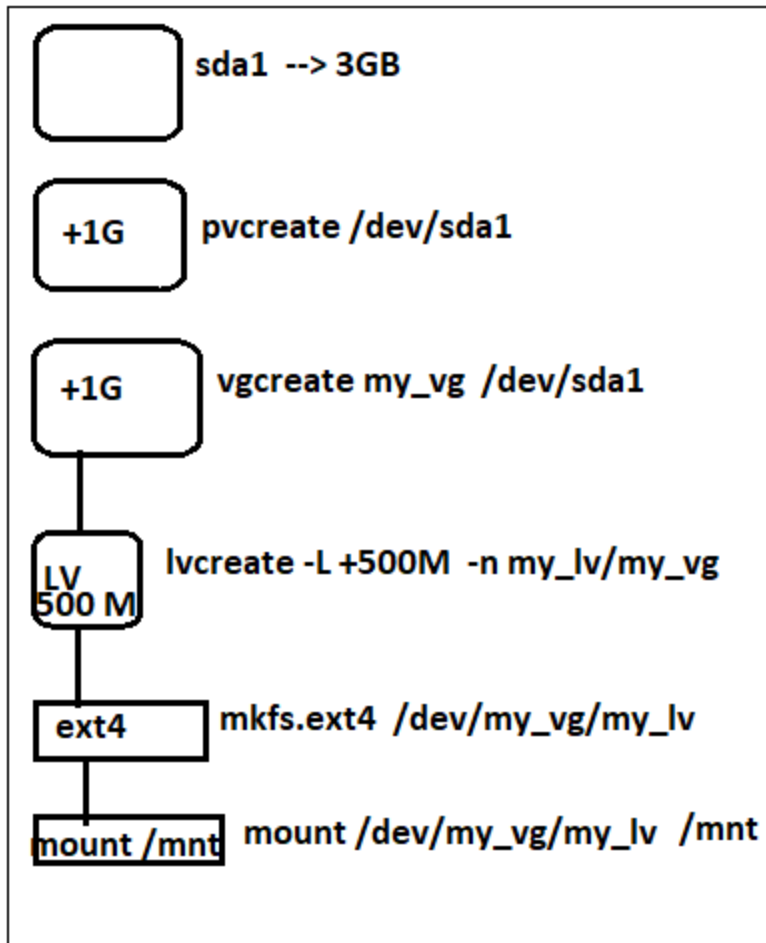
pvcreate /dev/sda1

vgcreate my_vg /dev/sda1

lvcreate -L +500M -n my_lv my_vg

mkfs.ext3 /dev/my_vg/my_lv

mount /dev/my_vg/my_lv /mnt



/dev

```

[root@manish ~]# cd /dev/
[root@manish dev]# ls
autofs          dm-0          hugepages     mapper        nvme0n1p2    rtc0          stderr        tty15         tty25         tty35         tty45         tty55         tty8           usbmon1        vcsa1         vcsu5
block           dm-1          hwrng         mcelog        nvme0n1p3    sda           stdin         tty16         tty26         tty36         tty46         tty56         tty9           usbmon2        vcsa2         vcsu6
bsg             dma_heap      initctl       mem           nvram         sda1          stdout        tty17         tty27         tty37         tty47         tty57         tty50          userfaultfd    vcsa3         vfio
bus            dmmidi        input         midi          port          sg0           tty           tty18         tty28         tty38         tty48         tty58         tty51          vcs         vcsa4         vga_arbiter
cdrom           dri           kmsg          queue         ppp           sg1           tty0          tty19         tty29         tty39         tty49         tty59         tty52          vcs1         vcsa5         vhci
char            fb0           log           net           ptmx          sg2           tty1          tty2          tty3          tty4          tty5          tty6          tty53          vcs2         vcsa6         vhost-net
console         fd            loop-control  ng0n1         pts           Sh            tty10         tty20         tty30         tty40         tty50         tty60         tty61          vcs3         vcsu         vhost-vsock
core            full          lp0           null          random        snapshot      tty11         tty21         tty31         tty41         tty51         tty61         uhid           vcs4         vcsu1         vmci
cpu             fuse          lp1           nvme0         rfcill        snd           tty12         tty22         tty32         tty42         tty52         tty62         uinput         vcs5         vcsu2         vsock
cpu_dma_latency hidraw0       lp2           nvme0n1       rhel          sr0           tty13         tty23         tty33         tty43         tty53         tty63         urandom        vcs6         vcsu3         zero
disk            hpet         lp3           nvme0n1p1     rtc           sri           tty14         tty24         tty34         tty44         tty54         tty64         usbmon0        vcsa         vcsu4

```

Partition with MBR

```

[root@manish ~]# fdisk /dev/sda

Welcome to fdisk (util-linux 2.37.4).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table.
Created a new DOS disklabel with disk identifier 0xc31997b1.

Command (m for help): n
Partition type
   p   primary (0 primary, 0 extended, 4 free)
   e   extended (container for logical partitions)
Select (default p): p
Partition number (1-4, default 1): 1
First sector (2048-10485759, default 2048):
Last sector, +/-sectors or +/-size(K,M,G,T,P) (2048-10485759, default 10485759): +1G

Created a new partition 1 of type 'Linux' and of size 1 GiB.
Partition #1 contains a ext3 signature.

Do you want to remove the signature? [Y]es/[N]o: y

The signature will be removed by a write command.

Command (m for help): p
Disk /dev/sda: 5 GiB, 5368709120 bytes, 10485760 sectors
Disk model: VMware Virtual S
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xc31997b1

Device      Boot Start      End Sectors Size Id Type
/dev/sda1                2048 2099199 2097152   1G 83 Linux

Filesystem/RAID signature on partition 1 will be wiped.

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.

```

Format and mount

```
[root@manish ~]# mkfs.ext3 /dev/sda1
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 262144 4k blocks and 65536 inodes
Filesystem UUID: 0c81bb74-7c69-4c7d-be77-8b9ce2875e30
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376

Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done

[root@manish ~]# mount /dev/sda1 /mnt
[root@manish ~]#
```

Check lsblk command

```
[root@manish ~]# lsblk
NAME                MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda                  8:0    0    5G  0 disk
└─sda1               8:1    0    1G  0 part /mnt
sr0                  11:0    1 112.5M  0 rom  /run/media/manish/CDROM
sr1                  11:1    1   9.8G  0 rom  /run/media/manish/RHEL-9-3-0-BaseOS-x86_64
nvme0n1             259:0    0   20G  0 disk
├─nvme0n1p1         259:1    0   600M  0 part /boot/efi
├─nvme0n1p2         259:2    0    1G  0 part /boot
└─nvme0n1p3         259:3    0  18.4G  0 part
    └─rhel-root      253:0    0  16.4G  0 lvm  /
        └─rhel-swap  253:1    0    2G  0 lvm  [SWAP]
```

Permanet mounting /etc/fstab

```
/dev/mapper/rhel-root / xfs defaults 0 0
UUID=78aa86d6-843d-4de5-b537-1c9aeae12252 /boot xfs defaults
UUID=E60B-FE4B /boot/efi vfat umask=0077,shortname=winnt 0 2
/dev/mapper/rhel-swap none swap defaults 0 0
/dev/sda1 /mnt ext3 defaults 0 0

[root@manish ~]# vim /etc/fstab
[root@manish ~]# mount -a
[root@manish ~]#
```

Partition with GPT

```
[root@manish ~]# gdisk /dev/sda
GPT fdisk (gdisk) version 1.0.7

Partition table scan:
  MBR: not present
  BSD: not present
  APM: not present
  GPT: not present

Creating new GPT entries in memory.

Command (? for help): n
Partition number (1-128, default 1): 1
First sector (34-10485726, default = 2048) or {+-}size{KMGTP}:
Last sector (2048-10485726, default = 10485726) or {+-}size{KMGTP}: +2G
Current type is 8300 (Linux filesystem)
Hex code or GUID (L to show codes, Enter = 8300): 8300
Changed type of partition to 'Linux filesystem'

Command (? for help): w

Final checks complete. About to write GPT data. THIS WILL OVERWRITE EXISTING
PARTITIONS!!

Do you want to proceed? (Y/N): y
OK; writing new GUID partition table (GPT) to /dev/sda.
The operation has completed successfully.
[root@manish ~]# █
```

Formate and mountng

```
[root@manish ~]# mkfs.ext4 /dev/sda1
mke2fs 1.46.5 (30-Dec-2021)
/dev/sda1 contains a ext3 file system
    last mounted on Fri Jul  4 23:07:33 2025
Proceed anyway? (y,N) y
Creating filesystem with 524288 4k blocks and 131072 inodes
Filesystem UUID: ac862060-a00d-4945-b719-d2d5237315c9
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

[root@manish ~]# mount /dev/sda1 /mnt
mount: (hint) your fstab has been modified, but systemd still uses
    the old version; use 'systemctl daemon-reload' to reload.
[root@manish ~]#
```

SWAP MEMORY

```
[root@manish ~]# fdisk /dev/sda
```

```
Welcome to fdisk (util-linux 2.37.4).
```

```
Changes will remain in memory only, until you decide to write them.
```

```
Be careful before using the write command.
```

```
Device does not contain a recognized partition table.
```

```
Created a new DOS disklabel with disk identifier 0xfcc5a88b.
```

```
Command (m for help): n
```

```
Partition type
```

```
  p   primary (0 primary, 0 extended, 4 free)
```

```
  e   extended (container for logical partitions)
```

```
Select (default p): p
```

```
Partition number (1-4, default 1): 1
```

```
First sector (2048-10485759, default 2048):
```

```
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-10485759, default 10485759): +2G
```

```
Created a new partition 1 of type 'Linux' and of size 2 GiB.
```

```
Command (m for help): t
```

```
Selected partition 1
```

```

Selected partition 1
Hex code or alias (type L to list all): l

00 Empty                24 NEC DOS               81 Minix / old Lin     bf Solaris
01 FAT12                 27 Hidden NTFS Win     82 Linux swap / So    c1 DRDOS/sec (FAT-
02 XENIX root            39 Plan 9               83 Linux               c4 DRDOS/sec (FAT-
03 XENIX usr              3c PartitionMagic      84 OS/2 hidden or     c6 DRDOS/sec (FAT-
04 FAT16 <32M            40 Venix 80286          85 Linux extended     c7 Syrix
05 Extended              41 PPC PReP Boot       86 NTFS volume set    da Non-FS data
06 FAT16                 42 SFS                  87 NTFS volume set    db CP/M / CTOS / .
07 HPFS/NTFS/exFAT       4d QNX4.x              88 Linux plaintext    de Dell Utility
08 AIX                   4e QNX4.x 2nd part     8e Linux LVM          df BootIt
09 AIX bootable          4f QNX4.x 3rd part     93 Amoebe             e1 DOS access
0a OS/2 Boot Manag       50 OnTrack DM          94 Amoebe BBT         e3 DOS R/O
0b W95 FAT32             51 OnTrack DM6 Aux     9f BSD/OS             e4 SpeedStor
0c W95 FAT32 (LBA)       52 CP/M                a0 IBM Thinkpad hi   ea Linux extended
0e W95 FAT16 (LBA)       53 OnTrack DM6 Aux     a5 FreeBSD            eb BeOS fs
0f W95 Ext'd (LBA)       54 OnTrackDM6          a6 OpenBSD            ee GPT
10 OPUS                  55 EZ-Drive            a7 NeXTSTEP           ef EFI (FAT-12/16/
11 Hidden FAT12          56 Golden Bow          a8 Darwin UFS         f0 Linux/PA-RISC b
12 Compaq diagnost       5c Priam Edisk         a9 NetBSD             f1 SpeedStor
14 Hidden FAT16 <3       61 SpeedStor           ab Darwin boot        f4 SpeedStor
16 Hidden FAT16          63 GNU HURD or Sys     af HFS / HFS+         f2 DOS secondary
17 Hidden HPFS/NTF       64 Novell Netware      b7 BSDI fs            fb VMware VMFS
18 AST SmartSleep        65 Novell Netware      b8 BSDI swap          fc VMware VMKCORE
1b Hidden W95 FAT3       70 DiskSecure Mult    bb Boot Wizard hid   fd Linux raid auto
1c Hidden W95 FAT3       75 PC/IX               bc Acronis FAT32 L   fe LANstep
1e Hidden W95 FAT1       80 Old Minix           be Solaris boot       ff BBT

Aliases:
linux      - 83
swap      - 82
extended  - 05
uefi      - EF
raid      - FD
lvm       - 8E
linuxex   - 85

Hex code or alias (type L to list all): 82
Changed type of partition 'Linux' to 'Linux swap / Solaris'.

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.

```

MKSWAP AND SWAPON

```

[root@manish ~]# mkswap /dev/sda1
mkswap: /dev/sda1: warning: don't erase bootbits sectors
        (gpt partition table detected). Use -f to force.
Setting up swapspace version 1, size = 2 GiB (2147479552 bytes)
no label, UUID=6dd2d26c-f788-4026-818b-795a4c614005
[root@manish ~]# swapon /dev/sda1
[root@manish ~]# █

```

BEFORE

```
[root@manish ~]# free -h
```

	total	used	free	shared	buff/cache	available
Mem:	1.7Gi	1.3Gi	114Mi	17Mi	422Mi	369Mi
Swap:	2.0Gi	94Mi	1.9Gi			

```
[root@manish ~]#
```

AFTER SWAPON

```
[root@manish ~]# free -h
```

	total	used	free	shared	buff/cache	available
Mem:	1.7Gi	1.3Gi	107Mi	17Mi	422Mi	362Mi
Swap:	4.0Gi	94Mi	3.9Gi			

```
[root@manish ~]#
```

Vim /etc/fstab

```
/dev/mapper/rhel-root / xfs defaults 0 0
UUID=78aa86d6-843d-4de5-b537-1c9aee12252 /boot xfs defaults 0 0
UUID=E60B-FE4B /boot/efi vfat umask=0077,shortname=winnt 0 2
/dev/mapper/rhel-swap none swap defaults 0 0
/dev/sda1 swap swap defaults 0 0
```

Create LVM

CREATE sda1


```

[root@manish ~]#
[root@manish ~]# fdisk /dev/sda

Welcome to fdisk (util-linux 2.37.4).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table.
Created a new DOS disklabel with disk identifier 0x34f2a376.

Command (m for help): n
Partition type
   p   primary (0 primary, 0 extended, 4 free)
   e   extended (container for logical partitions)
Select (default p):

Using default response p.
Partition number (1-4, default 1):
First sector (2048-10485759, default 2048):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-10485759, default 10485759): +4G

Created a new partition 1 of type 'Linux' and of size 4 GiB.
Partition #1 contains a swap signature.

Do you want to remove the signature? [Y]es/[N]o: y

The signature will be removed by a write command.

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.

```

pvccreate, vgcreate, lvcreate & formate, mounting

```

[root@manish ~]# pvcreate /dev/sdal
Physical volume "/dev/sdal" successfully created.
[root@manish ~]# vgcreate my_vg /dev/sdal
WARNING: adding device /dev/sdal with idname naa.5000c29927421f5e which is already used for missing device.
Volume group "my_vg" successfully created
[root@manish ~]# lvcreate -L +1G -n my_lv my_vg
Logical volume "my_lv" created.
[root@manish ~]# mkfs.ext4 /dev/my_vg/my_lv
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 262144 4k blocks and 65536 inodes
Filesystem UUID: 9d82ad72-44d0-4b95-8955-b6b238c24dfc
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376

Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done

[root@manish ~]# mount /dev/my_vg/my_lv /mnt
mount: (hint) your fstab has been modified, but systemd still uses
        the old version; use 'systemctl daemon-reload' to reload.
[root@manish ~]#

```

Check lsblk command

```
[root@manish ~]# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
├─sda         8:0    0   5G  0 disk
│ └─sda1      8:1    0   4G  0 part
│   └─my_vg-my_lv 253:2  0   1G  0 lvm  /mnt
├─sr0        11:0    1 112.5M  0 rom  /run/media/manish/CDROM
├─sr1        11:1    1   9.8G  0 rom  /run/media/manish/RHEL-9-3-0-BaseOS-x86_64
├─nvme0n1     259:0    0   20G  0 disk
│ └─nvme0n1p1 259:1    0   600M  0 part  /boot/efi
│   └─nvme0n1p2 259:2    0    1G  0 part  /boot
│     └─nvme0n1p3 259:3    0  18.4G  0 part
│       └─rhel-root 253:0    0  16.4G  0 lvm  /
│         └─rhel-swap 253:1    0    2G  0 lvm  [SWAP]
└─
```

Vim /etc/fstab

```
#
# /etc/fstab
# Created by anaconda on Tue Dec  3 09:24:54 2024
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
/dev/mapper/rhel-root / xfs defaults 0 0
UUID=78aa86d6-843d-4de5-b537-1c9aeae12252 /boot xfs defaults 0 0
UUID=E60B-FE4B /boot/efi vfat umask=0077,shortname=winnt 0 2
/dev/mapper/rhel-swap none swap defaults 0 0
/dev/my_vg/my_lv /mnt ext4 defaults 0 0
```

lvremove,vgremove,pvremove command

```
[root@manish ~]# umount /mnt
[root@manish ~]# lvremove /dev/my_vg/my_lv
Do you really want to remove active logical volume my_vg/my_lv? [y/n]: y
Logical volume "my_lv" successfully removed.
[root@manish ~]# vgremove /dev/my_vg
Volume group "my_vg" successfully removed
[root@manish ~]# pvremove /dev/sda1
Labels on physical volume "/dev/sda1" successfully wiped.
[root@manish ~]#
```

Lvresize

Lvresize -rL +80M /dev/my_vg/my_lv= increase

Lvresize -rL 40M /dev/my_vg/my_lv= decrease

```
[root@manish ~]# lvresize -rL +60M /dev/my_vg/my_lv
Size of logical volume my_vg/my_lv changed from 52.00 MiB (13 extents) to 112.00 MiB (28 extents).
File system ext4 found on my_vg/my_lv mounted at /mnt.
Extending file system ext4 to 112.00 MiB (117440512 bytes) on my_vg/my_lv...
resize2fs /dev/my_vg/my_lv
resize2fs 1.46.5 (30-Dec-2021)
Filesystem at /dev/my_vg/my_lv is mounted on /mnt; on-line resizing required
old_desc_blocks = 1, new_desc_blocks = 1
The filesystem on /dev/my_vg/my_lv is now 114688 (1k) blocks long.

resize2fs done
Extended file system ext4 on my_vg/my_lv.
Logical volume my_vg/my_lv successfully resized.
[root@manish ~]#
Logical volume my_vg/my_lv successfully resized.
[root@manish ~]# lvresize -rL 30M /dev/my_vg/my_lv
Rounding size to boundary between physical extents: 32.00 MiB.
File system ext4 found on my_vg/my_lv mounted at /mnt.
File system size (112.00 MiB) is larger than the requested size (32.00 MiB).
File system reduce is required using resize2fs.
File system unmount is needed for reduce.
File system fsck will be run before reduce.
Continue with ext4 file system reduce steps: unmount, fsck, resize2fs? [y/n]:y
Reducing file system ext4 to 32.00 MiB (33554432 bytes) on my_vg/my_lv...
unmount /mnt
unmount done
e2fsck /dev/my_vg/my_lv
/dev/my_vg/my_lv: 11/26656 files (0.0% non-contiguous), 12351/114688 blocks
e2fsck done
resize2fs /dev/my_vg/my_lv 32768k
resize2fs 1.46.5 (30-Dec-2021)
Resizing the filesystem on /dev/my_vg/my_lv to 32768 (1k) blocks.
The filesystem on /dev/my_vg/my_lv is now 32768 (1k) blocks long.

resize2fs done
remount /dev/my_vg/my_lv /mnt
mount: (hint) your fstab has been modified, but systemd still uses
the old version; use 'systemctl daemon-reload' to reload.
remount done
Reduced file system ext4 on my_vg/my_lv.
Size of logical volume my_vg/my_lv changed from 112.00 MiB (28 extents) to 32.00 MiB (8 extents).
Logical volume my_vg/my_lv successfully resized.
[root@manish ~]#
```

Note: The partition can only be resized if it is formatted from ext4 or xfs.

et4: With this you can increase and decrease the size of LV.

Xfs: only LVM Increase no Decrease.

Explain /etc/fstab

6 fields in this file

1. Path/UUID --> for eg. /dev/sda1
2. Mount point --> for eg. /mnt
3. Filesystem --> for eg. Ext4
4. Permission --> defaults
5. Dump frequency --> 0
6. FSCK --> file system consistency check --> 0

/dev/my_vg/my_lv	/mnt	ext4	defaults	0	0
path	mount point	file system	permission	dump frequency	FSCK

Thank You 😊

COMPLETE STORAGE



😊 M.S

